

NURa Hand-in 1

Name (Student number)

Date

Write a short description of your solution in each section. Explain your choices. Draw the conclusions from your results.

You can refer to selected lines of code where you think it is necessary. Make sure the line numbers you reference are after “black” has been run on your files.

Don't forget to describe your figures and mention what conclusions you draw from them!

1 Simulating the solar system

1.1 (1a)

Make a plot showing the (x, y) positions and another showing the (x, z) positions for all the objects at the “current time” defined in the example.

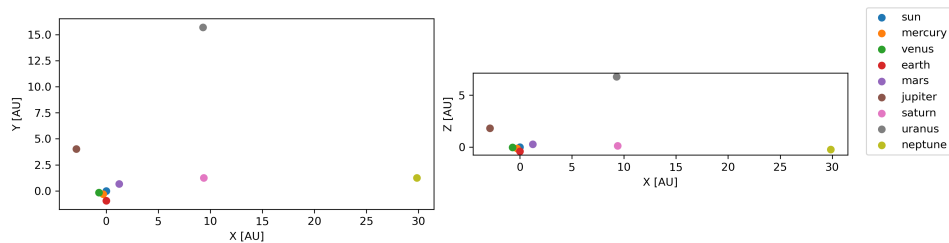


Figure 1: Initial positions of the objects in the solar system in the x-y and x-z planes.

1.2 (1b)

Show the orbits of all objects in the x - y plane in a suitable reference frame, as well as a plot of time versus position in the z plane. Explain whether leapfrog is a suitable choice of an algorithm, and why.

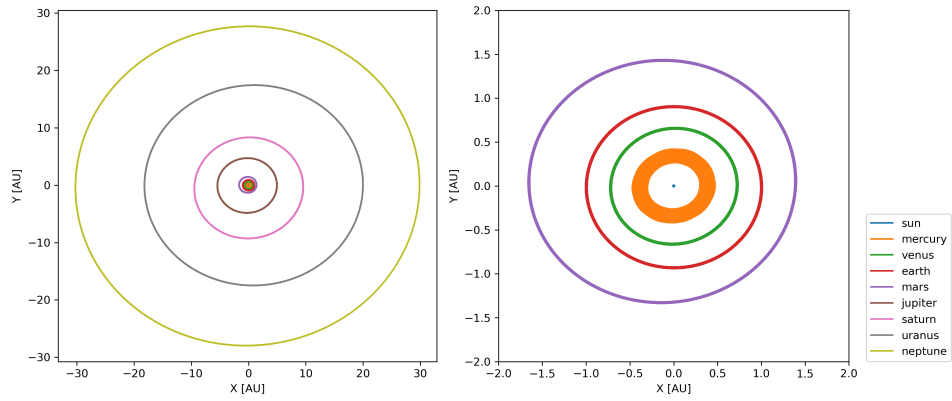


Figure 2: Orbits of the objects in the x-y plane using the leapfrog integrator.

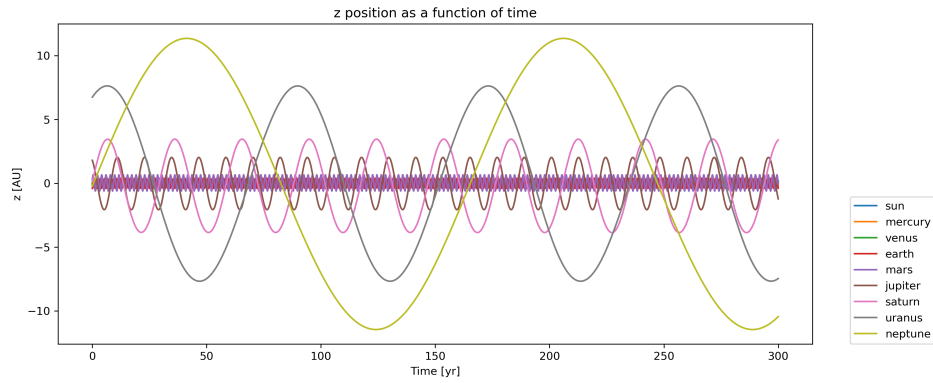


Figure 3: Position in the z plane as a function of time for all objects using the leapfrog integrator.

1.3 (1c)

Explain the integration algorithm you used. What are the differences compared to the leapfrog method?

Make the same plots that you did in (b), as well as a plot of the difference in the bodies' x positions versus time between the leapfrog method and the other method. Discuss your results.

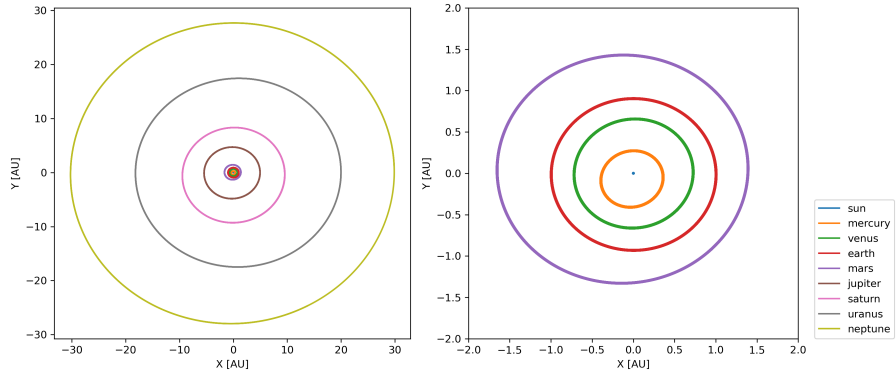


Figure 4: Orbits of the objects in the x-y plane using another integration method.

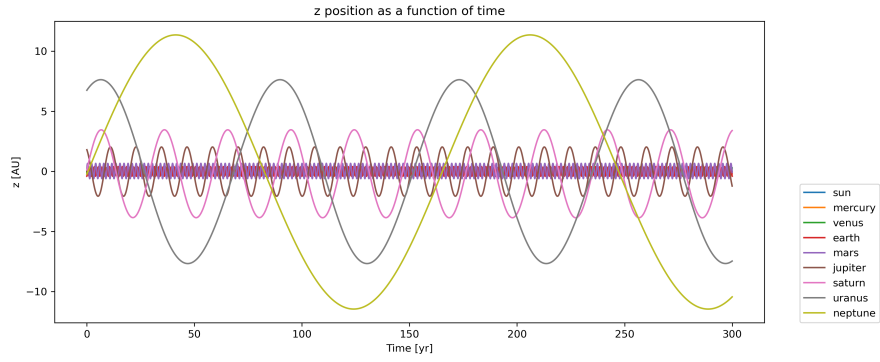


Figure 5: Position in the z plane as a function of time for all objects using another integration method.

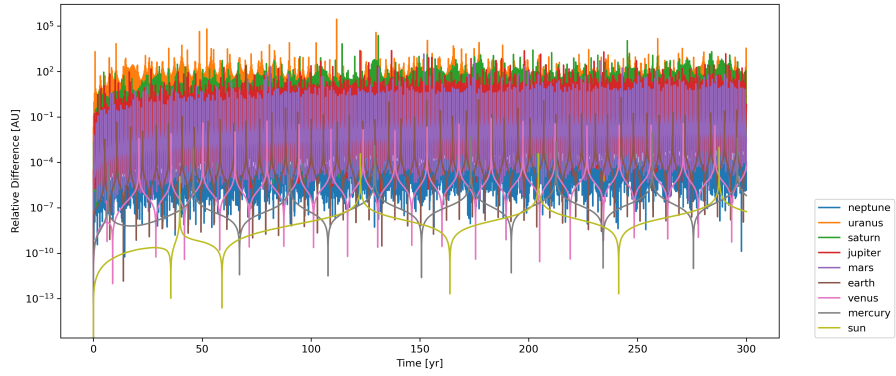


Figure 6: Difference in the x positions of the bodies as a function of time between the leapfrog method and another integration method.

2 Calculating potentials in different ways

2.1 (2a)

Describe how you constructed your octree.

Plot the mass distribution for four slices at different depth levels, namely the first four slices in xi at levels 3, 5 and 7.

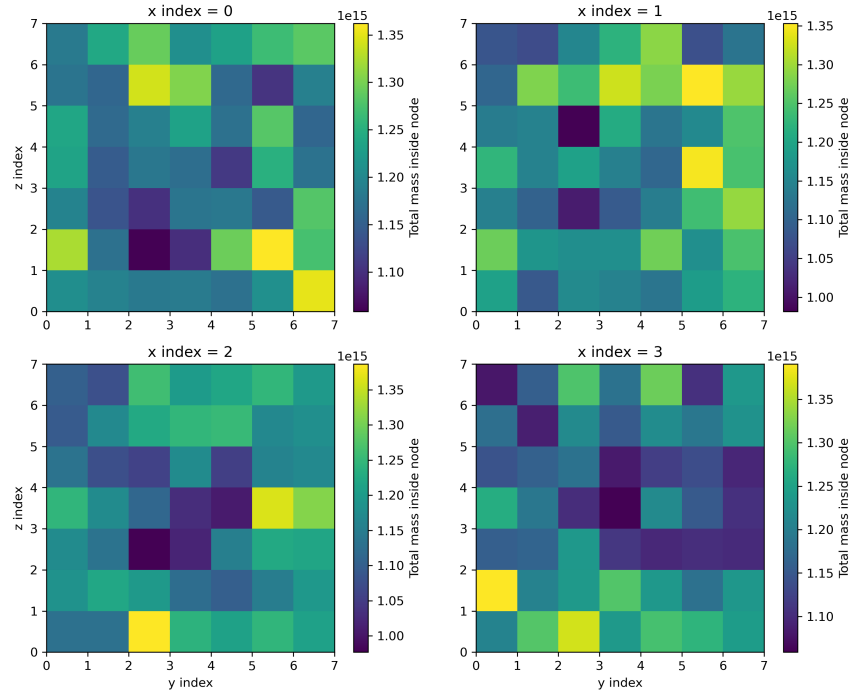


Figure 7: Mass distribution for the first four slices in xi at level 3 of the octree.

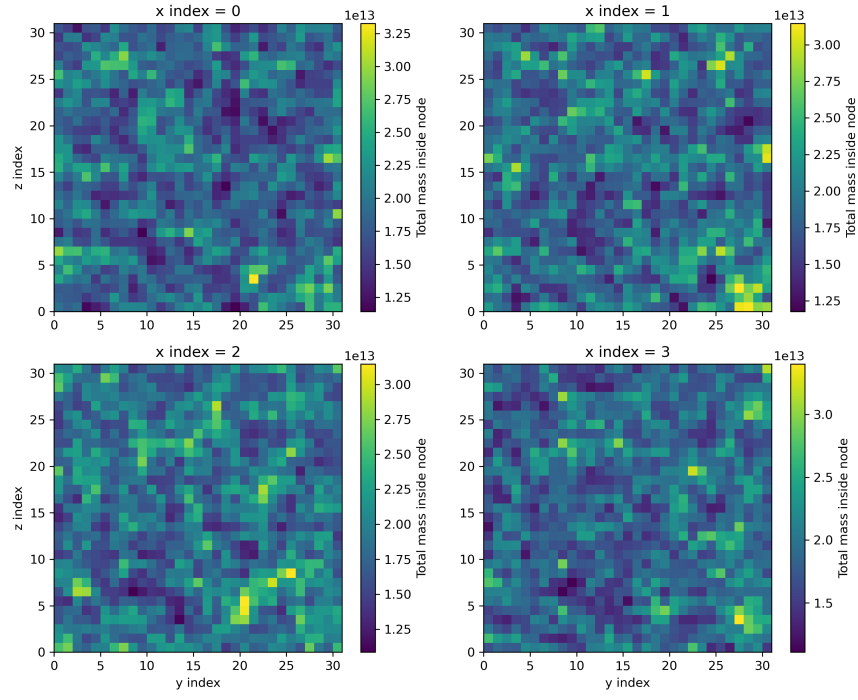


Figure 8: Mass distribution for the first four slices in xi at level 5 of the octree.

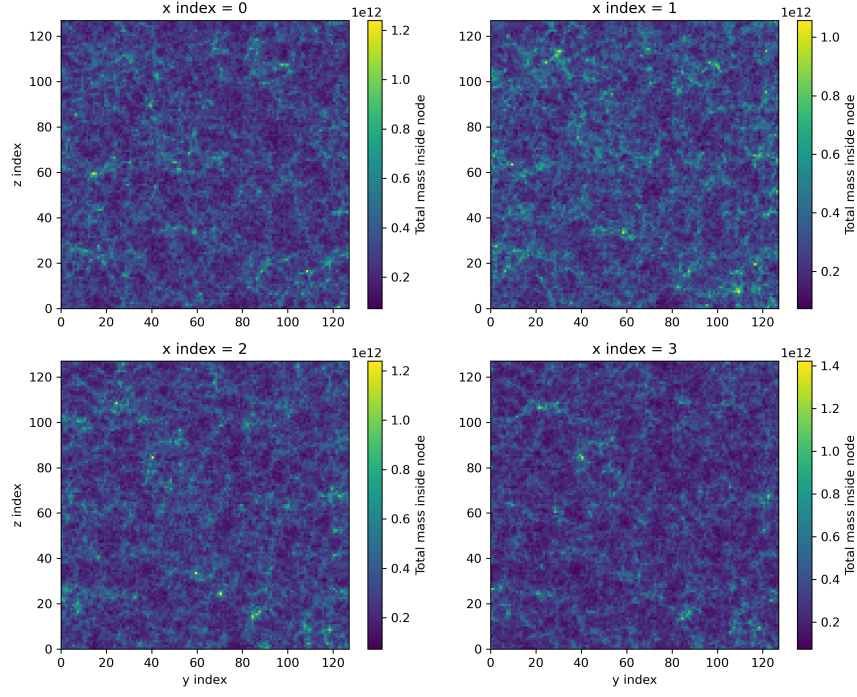


Figure 9: Mass distribution for the first four slices in xi at level 7 of the octree.

2.2 (2b)

Describe how you computed the potential and implemented fft.

Plot the results for different slices.

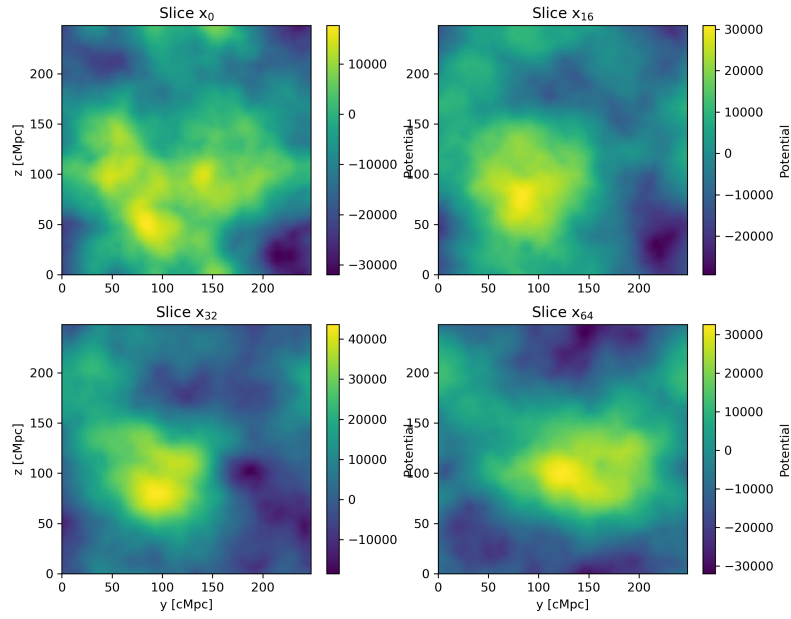


Figure 10: Potential computed using the FFT method for different slices.

Discuss the results.

3 Spiral and elliptical galaxies

3.1 (3a)

Plot the distributions of the rescaled features, and save them to a text file; print the features for the first 10 objects in your report (do not hardcode them into the report, but show the first 10 lines of the text file).

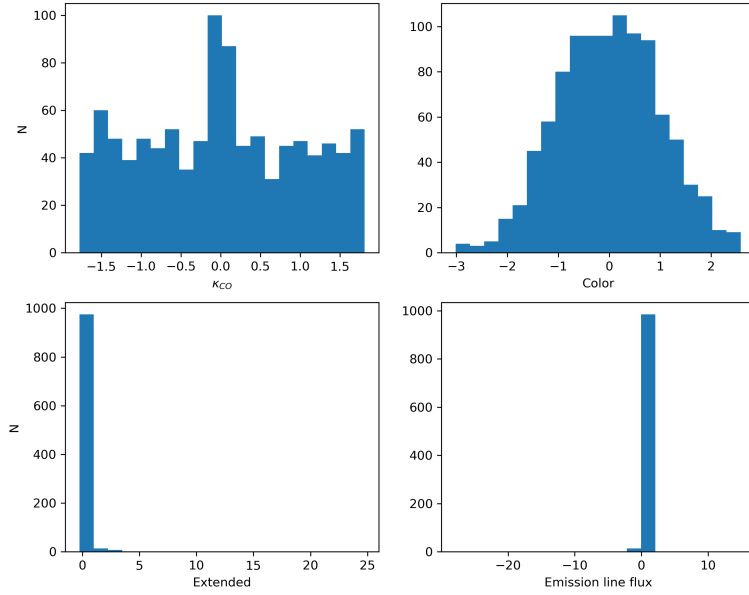


Figure 11: Distributions of the rescaled features for the spiral and elliptical galaxies.

```

1 # kappa_CO Color Extended Emission_line_flux
2 -1.581862133006100457e+00 -6.612307828918272123e-03 -3.475533387671270874
   e-02 4.149921715612883179e-03
3 1.574559067587912864e+00 -7.944354060046805444e-01 2.031450234565141122e
   -01 1.605487869858178776e-02
4 1.531990475165861731e+00 8.573405014974259108e-01 -2.079754178048099089e
   -01 4.166315770216804520e-02
5 1.297215471156621280e+00 1.226267345173265966e+00 1.858865005815530658e
   -01 6.825469423304415632e-04
6 5.181259591131053277e-01 8.257714425578112083e-01 -2.002589858418063196e
   -01 1.555766189074685689e-02
7 -1.647524013432830392e+00 -2.909640578232330355e-01 -8.488083597139740866
   e-02 1.855013775209116709e-03
8 -1.312495960360360536e+00 -9.756536637063872863e-01 6.454576379671399244e
   -02 6.772650820885567705e-03
9 9.418303994505568266e-02 -9.715157240517418025e-01 -1.525183484703762915e
   -01 7.745013475795202093e-03
10 -7.308293662245411992e-01 -1.520148623547937117e+00 -1.654403149798441552
   e-01 1.175440493844443976e-02
11 -1.198035514238126931e+00 -7.748995242729109778e-01 -1.553888673156534339
   e-01 9.596892572192079404e-03

```


3.2 (3b)

Explain how you implemented logistic regression. Plot the evolution of your cost function with the number of iterations for different pairs of features. Explain how does it change depending on the feature pair.

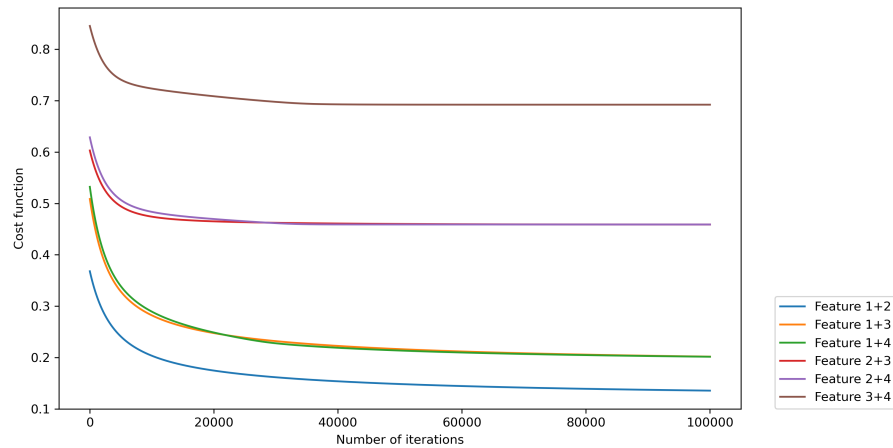


Figure 12: Evolution of the cost function with the number of iterations for different pairs of features in the logistic regression model.

3.3 (3c)

Print the values of true/false positives/negatives, as well as the F1 score.

```
1 True Positives: 474
2 False Positives: 25
3 True Negatives: 475
4 False Negatives: 26
5 F1 Score: 0.9489
```

Plot the two features against each other and indicate the decision boundary and discuss your results.

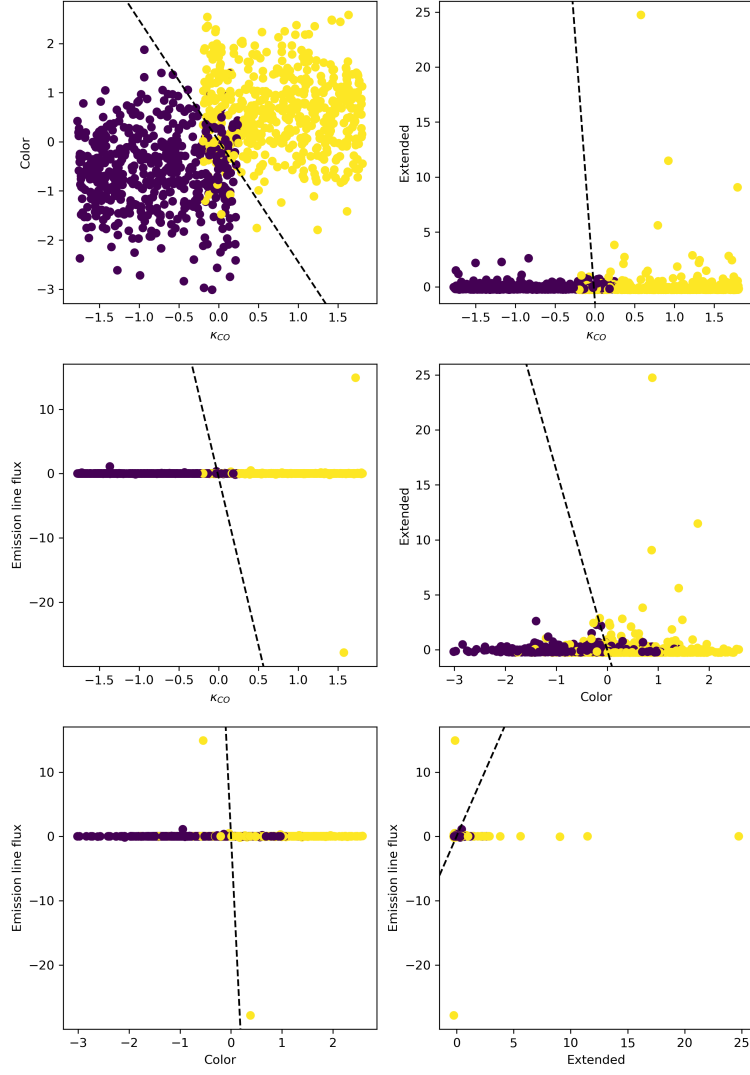


Figure 13: Plot of the pairs of two features against each other with the decision boundary.